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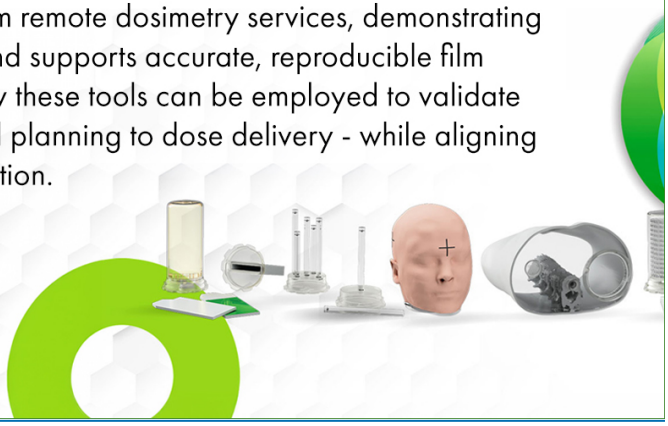
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Percentile-based probabilistic optimization for systematic and random uncertainties in radiation therapy

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Abstract

Objective. Geometric uncertainty can degrade treatment quality in radiation therapy. While margins and robust optimization mitigate these effects, they provide only implicit control over the probability of meeting the clinical goals. The goal of this work is to develop and evaluate a probabilistic optimization framework that explicitly targets a specified probability of clinical goal fulfillment over full treatment courses, while remaining computationally tractable. *Approach.* A probabilistic planning framework using a percentile-based optimization function was formulated. Systematic and random uncertainties were explicitly modeled over full treatment courses. To enable efficient optimization, treatment course doses were approximated using interpolation between a precomputed set of dose distributions. The framework was evaluated on a prostate case treated with volumetric-modulated arc therapy (VMAT) and a brain case treated with pencil beam scanning (PBS) proton therapy. Plans were compared to conventional margin-based and worst-case robust optimization using probabilistic evaluation with randomly sampled simulated treatment courses. *Main results.* For the prostate VMAT case, probabilistic optimization improved organ at risk (OAR) sparing while maintaining target coverage compared to margin-based planning, increasing average OAR goal fulfillment probability by 13.3 percentage points and reducing 90th percentile OAR doses by an average of 3.5 Gy. For the brain PBS case, probabilistic optimization improved target minimum dose passing probabilities (e.g. 84% vs 19% for D_{95}) and brainstem core maximum dose passing probability (56% vs 2%), while maintaining comparable or improved OAR sparing compared to worst-case optimization. *Significance.* Percentile-based probabilistic optimization enables explicit and interpretable control over clinical goal fulfillment probabilities. By combining full treatment course modeling with efficient approximate dose calculation, the proposed framework provides a practical approach for incorporating probabilities into treatment planning.

1. Introduction

Geometric uncertainty, such as patient setup and range uncertainty, can degrade radiation therapy treatment quality. Robust optimization has become the standard approach to mitigate these effects in proton therapy, and is increasingly being adopted for photon therapy in situations where the static dose cloud approximation is inaccurate and margins fail to provide adequate protection (Archibald-Heeren *et al* 2017, Zhang *et al* 2018, Kaplan *et al* 2022, Sterpin *et al* 2024).

Margin recipes often aim for a specified probability of target coverage under systematic and random setup errors (Van Herk *et al* 2000). Conventional worst-case and expected value optimization frameworks (Unkelbach *et al* 2009, Fredriksson *et al* 2011), which form the basis of most clinical

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implementations, enable implicit control over the probability through scenario selection—for example, the magnitudes of uncertainties to consider can be selected in accordance with the margin recipe (Sterpin *et al* 2024). However, neither margin-based planning nor conventional robust optimization explicitly optimizes the probability of goal fulfillment.

Probabilistic planning has been explored to provide explicit control over probabilities. Löf *et al* (1995) simulated full treatment courses under systematic and random errors for 1D phantoms and maximized the probability of complication-free tumor control. Witte *et al* (2007), Gordon *et al* (2010) and Bohoslavsky *et al* (2013) extended these ideas to clinical cases, using sampling to handle systematic errors and dose or fluence blurring for random errors. Witte *et al* focused on tumor control and normal tissue complication probability, whereas Gordon *et al* and Bohoslavsky *et al* optimized for a selected probability of goal fulfillment. Tilly *et al* (2019) optimized the conditional value-at-risk (CVaR), iteratively adjusting its percentile parameter until the desired goal fulfillment probability was achieved. In brachytherapy, Kennedy *et al* (2025) used genetic programming to optimize the probability of fulfilling the clinical goals.

Recent studies have considered full treatment course simulation for probabilistic evaluation (Tilly and Ahnesjö 2015, Souris *et al* 2019, Rojo-Santiago *et al* 2023, Olovsson *et al* 2025, de Jong *et al* 2025). In particular, Sterpin *et al* (2021) emphasize the importance of statistically sound robustness evaluation, showing that full Monte Carlo simulation of treatment courses can enable more accurate trade-offs than conventional evaluation strategies that use only a few scenarios. They also highlight the need for fast treatment course dose calculation to make comprehensive robustness evaluation practical. Examples of such fast approaches include methods that precompute a set of dose distributions and combine them by interpolation to simulate full treatment courses (Rojo-Santiago *et al* 2023, de Jong *et al* 2026).

Building on the work by Bohoslavsky *et al* (2013), we propose a probabilistic optimization approach that explicitly models both systematic and random uncertainties over full treatment courses. The framework directly targets a specified probability of meeting the planning goals using percentile-based optimization functions. To make the optimization computationally tractable, we adapt a recently proposed fast robustness evaluation method (de Jong *et al* 2026) to the optimization setting, allowing efficient simulation of full treatment courses. We demonstrate the approach on a prostate case treated with volumetric-modulated arc therapy (VMAT) and a brain case treated with pencil beam scanning (PBS) proton therapy.

2. Methods

2.1. Framework

In essence, the proposed probabilistic approach evaluates the candidate treatment plan over many simulated treatment course scenarios and optimizes a percentile-based objective over these scenarios.

The framework combines three main components: (i) modeling of treatment course scenarios to represent systematic and random uncertainties over full treatment courses, (ii) a percentile-based optimization function that enables direct control over the probability of clinical goal fulfillment, and (iii) an efficient scenario dose approximation method that makes the approach computationally tractable. These components are described in sections 2.1.1, 2.1.2, and 2.1.3, respectively.

In addition, we discuss how multiple objectives can be coupled through scenario selection (section 2.1.4) and formulate the resulting optimization problem (section 2.1.5).

An overview of the framework is illustrated in figure 1.

2.1.1. Treatment course scenarios

We model a treatment course as a sequence of fractions affected by both systematic and random errors. Considering setup uncertainty only, a treatment course scenario (or just 'scenario' for short) s is represented by a setup shift for each fraction, composed of a systematic component shared across fractions and a random per-fraction component:

$$s = \left(\Delta^{\text{syst}} + \Delta_1^{\text{rand}}, \ldots, \Delta^{\text{syst}} + \Delta_{n_\text{f}}^{\text{rand}}\right),$$

where Δ^{syst} denotes the systematic setup error, Δ_i^{rand} the random setup error at fraction i, and n_f the number of fractions. The total dose $d(x;s)$ in scenario s for variables x (e.g. spot weights for protons and multi-leaf collimator leaf positions, dose rate, and gantry speed for photons) is given by

$$d(x;s) = \sum_{i=1}^{n_\text{f}} d_\text{f}\left(x; \Delta^{\text{syst}} + \Delta_i^{\text{rand}}\right),$$

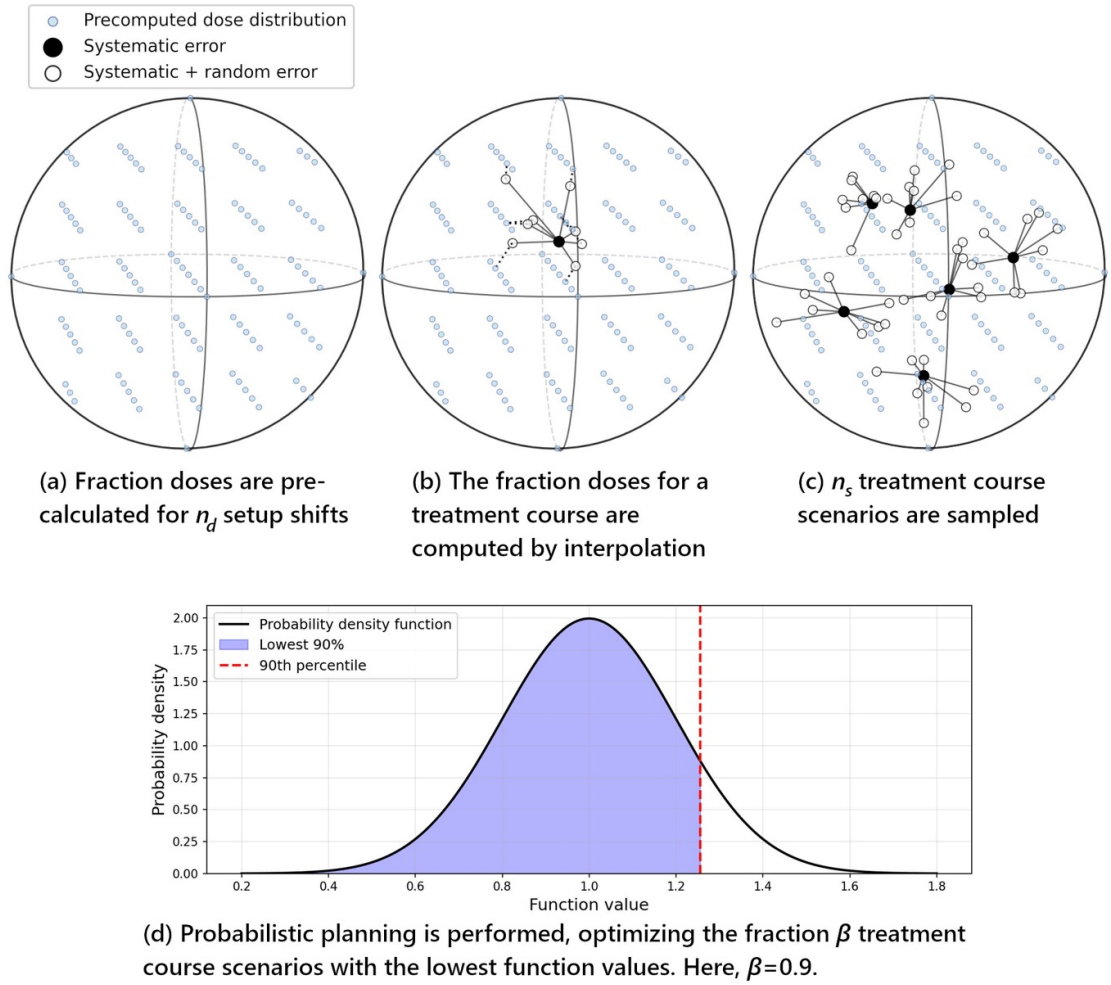


Figure 1. Overview of the proposed probabilistic optimization framework with approximate treatment course scenario dose calculation. (a) For the current variable values, fraction doses are precalculated on a grid of setup shifts within a specified range represented by the outer sphere. (b) A treatment course scenario is generated by sampling a systematic setup error and combining it with per-fraction random errors, and its constituent fraction doses are computed by interpolation in the grid (dotted lines indicate the nearest neighbors). (c) Multiple treatment course scenarios are sampled, and their doses computed by interpolation in the grid, to form an empirical distribution of scenario doses. (d) The function value of the objective is computed for each of the treatment course scenario dose distributions, and the percentile-trimmed power mean of these values is computed, enabling probabilistic optimization targeting the fraction β of best function values over the treatment course scenarios, i.e. the 100β-percentile.

where $d_f(x;\Delta)$ is the dose distribution from one fraction evaluated for variables x and setup error Δ. The formulation can easily be extended to include other uncertainty sources, such as range uncertainty.

2.1.2. Percentile-based optimization functions

To enable direct control over the probability of goal fulfillment, we evaluate the optimization functions over a distribution of treatment course scenarios and aggregate the results using a percentile-based risk measure. The method focuses the optimization on the fraction β of best-performing scenarios, corresponding to a target probability of goal fulfillment.

More specifically, we introduce a *percentile-trimmed power mean*, which extends earlier work by Bohoslavsky *et al* (2013). While their approach averages over the best-performing scenarios, the present formulation allows a smooth transition from such averaging towards directly optimizing a percentile (value-at-risk). The construction proceeds in three steps: (i) evaluate the optimization function for each scenario, (ii) select the fraction β of best-performing scenarios, and (iii) aggregate these values using a power mean.

We now give a formal definition. Let $f(d(x;S))$ denote an optimization function evaluated on the dose under scenario S, where the random variable S takes the value s from the set Ω of scenarios with

probability p_s. Define the function values for each scenario by

$$v_s = f(d(x;s)), \qquad s \in \Omega.$$

Order the values $v_{(1)} \leqslant \ldots \leqslant v_{(n_s)}$ with corresponding probabilities $p_{(i)}$, where n_s is the number of scenarios in Ω, and introduce cumulative probabilities

$$c_j = \sum_{i=1}^{j} p_{(i)}, \qquad j = 1, \ldots, n_s, \qquad c_0 = 0.$$

For a target percentile level 100β, $\beta \in (0,1]$, we identify the smallest index k such that $c_{k-1} < \beta \leqslant c_k$. This determines the fraction β of best-performing scenarios. In case of ties, we define

$$p_{\text{tie}} = \sum_{i:v_{(i)}=v_{(k)}} p_{(i)}$$

as the probability mass of the tie group with function values equal to $v_{(k)}$. Let t be the smallest index of that tie group.

From these quantities, we construct the trimmed distribution supported on the fraction β best scenarios and renormalize to get the conditional probabilities

$$\tilde{p}_{(i)} = \begin{cases} \frac{p_{(i)}}{\beta}, & v_{(i)} < v_{(k)}, \\ \frac{p_{(i)}}{\beta}\frac{\beta - c_{t-1}}{p_{\text{tie}}}, & v_{(i)} = v_{(k)}, \\ 0 & \text{otherwise.} \end{cases}$$

The percentile-trimmed power mean with power $a \neq 0$ at percentile 100β is then

$$Q_a^{\beta}(f(d(x;S))) = \left(\sum_{i=1}^{n_s} \tilde{p}_{(i)} v_{(i)}^a \right)^{1/a}.$$

Intuitively, this corresponds to averaging f over the best-performing fraction β of scenarios, while the parameter a controls how strongly the worst scenarios within this subset are emphasized. For $a = 1$, the measure reduces to the conditional expected value over the selected scenarios, whereas as $a \to \infty$, it approaches the 100β-percentile (value-at-risk). Intermediate values of a provide a smooth transition between these behaviors.

For $a \neq 1$, the function Q_a^{β} requires f to be nonnegative. Alternative percentile approximations, such as the log-sum-exp applied to the conditional distribution, could be used when negative values are present.

2.1.3. Treatment course scenario dose approximation

Simulating full treatment courses involves many dose calculations: for n_f fractions and n_s scenarios, $n_f \times n_s$ dose calculations are required per iteration. To reduce computational cost, we adopt the interpolation strategy from Fredriksson *et al* (2021), extended to handle both systematic and random errors by de Jong *et al* (2026). This approach drastically reduces the computational burden and has been shown to preserve adequate accuracy for robustness evaluation (de Jong *et al* 2026).

In each iteration, for the current optimization variables x, we precompute dose distributions $\hat{d}_j(x)$, $j = 1, \ldots, n_d$, on a regular 3D grid for setup shifts that are at most a distance R from the origin, with R selected to cover most of the uncertainty. The fraction dose $d_f(x;\Delta)$ for an arbitrary shift Δ is then approximated by

$$d_f(x;\Delta) = \sum_{j=1}^{n_d} \alpha_j(\Delta)\, \hat{d}_j(x),$$

where $\alpha_j(\Delta)$ are nonnegative interpolation weights, which can be selected using, e.g. nearest neighbor interpolation, inverse distance weighting, or trilinear interpolation.

The accumulated dose $d(x;s)$ over a treatment course is

$$d(x;s) = \sum_{i=1}^{n_f} \sum_{j=1}^{n_d} \alpha_j\left(\Delta^{\text{syst}} + \Delta_i^{\text{rand}}\right) \hat{d}_j(x) = \sum_{j=1}^{n_d} \gamma_j(s)\, \hat{d}_j(x), \qquad \gamma_j(s) = \sum_{i=1}^{n_f} \alpha_j\left(\Delta^{\text{syst}} + \Delta_i^{\text{rand}}\right).$$

This reduces the number n_d of dose calculations to the number of precomputed grid points. When the grid spacing is $R/2$, $R/3$, or $R/4$ and points outside the sphere of radius R are omitted, n_d becomes respectively 33, 123, or 257. Computing treatment course scenario doses is then a matter of weighted summation of these doses.

2.1.4. Correlation

Multiple optimization functions can either be handled independently or jointly. If two functions f and g are optimized using separate percentile functions,

$$Q_a^\beta(f) + Q_a^\beta(g),$$

then the optimization aims for a plan where each function has a low value in a fraction β of the scenarios, but this subset of scenarios may differ between the two functions. This is analogous to objective-wise worst-case in conventional robust optimization, where each objective is evaluated under its own worst-case scenario (Fredriksson and Bokrantz 2014).

Alternatively, optimization of the combined function,

$$Q_a^\beta(f+g),$$

aims for a low value of the sum of the functions within the same subset of scenarios, effectively coupling their probabilities. This is analogous to composite worst-case optimization, but with the difference that the coupling is defined by a percentile-based scenario selection rather than by an extremal worst-case.

This distinction determines whether clinical goals are required to be fulfilled simultaneously within the same treatment scenarios, or individually across different subsets of scenarios.

Another option is to choose a dominant percentile function and reuse its scenario ordering for correlated percentile functions.

2.1.5. Optimization formulation

The overall optimization problem, taking correlation into account in the objective, is

$$\begin{aligned} \underset{x\in X}{\text{minimize}} \quad & Q_a^\beta\left(\sum_{i\in O_\mathrm{prob}} w_i f_i(d(x;S))\right) + \sum_{i\in O_\mathrm{nom}} w_i f_i(d(x;s_\mathrm{nom})) \\ \text{subject to} \quad & Q_a^\beta(f_i(d(x;S))) \leqslant 0, \quad i\in C_\mathrm{prob}, \\ & f_i(d(x;s_\mathrm{nom})) \leqslant 0, \quad i\in C_\mathrm{nom}, \end{aligned}$$

where x denotes the variables, X is the set of feasible variable values, O_prob and O_nom are sets indexing the objective functions to be treated, respectively, probabilistically and nominally, and C_prob and C_nom are similar but for constraints, w_i is the importance weight of function f_i for $i \in O_\mathrm{prob} \cup O_\mathrm{nom}$, and s_nom denotes the nominal scenario. This formulation allows combining probabilistic and conventional planning objectives within the same optimization problem.

Here, correlation is only taken into account in the objective and not in the constraints. Such correlation could however be achieved by letting the percentile objective function dominate the scenario selection for all percentile constraint functions.

2.2. Computational study

Probabilistic optimization using the percentile-trimmed power mean was implemented in a research version of RayStation v2025 (RaySearch Laboratories, Sweden). It was applied to two patient cases and compared to conventional planning: a prostate case treated with VMAT where the conventional planning used margins, and a brain case treated with PBS protons where the conventional planning used worst-case optimization considering systematic range and setup uncertainty.

In the experiments, we used $\beta = 0.9$, corresponding to 90% probability of goal fulfillment, and $a = 8$, which is the power used in RayStation's approximation of worst-case optimization. Systematic and random setup errors were sampled from zero-mean isotropic 3D Gaussian distributions with standard deviations Σ and σ, respectively, and the systematic range uncertainty was sampled from a zero-mean 1D Gaussian with standard deviation Σ_r.

The VMAT optimization used a fast dose calculation algorithm based on singular value decomposition combined with collapsed cone computations at intermediate iterations, which were used to update the dose. The evaluation used collapsed cone calculations for each perturbed dose.

The PBS optimization used RayStation's approximate dose calculation, performing interpolation between the Monte Carlo calculated spots (Janson *et al* 2024), and the evaluation used full Monte Carlo calculation for each perturbed dose. A constant radiobiological effectiveness (RBE) factor of 1.1 was used.

The treatment course scenario dose approximation with spacing $R/2$ (33 grid points) was used in the optimization and $R/3$ (123 grid points) in the evaluation, and R was set to $3\sqrt{\Sigma^2+\sigma^2}$, based on the settings found to provide adequate accuracy in de Jong *et al* (2026). For the probabilistic PBS plan, range uncertainty was considered. To this end, the interval $[-2.6\Sigma_r, 2.6\Sigma_r]$ was discretized into 5 points in the optimization and 7 points in the evaluation, for a total of respectively $5\times 33 = 165$ and $7\times 123 = 861$ dose calculation points, which could be compared to the conventional robust optimization using $3\times 7 = 21$ dose calculation points. The interpolation weights α were selected by nearest neighbor interpolation. In optimization as well as evaluation, $n_s = 1000$ treatment courses were simulated, but using different random seeds and hence different treatment courses.

The computations were performed on a workstation with an Intel Core Ultra 9 285 K CPU, an NVIDIA RTX PRO 6000 Blackwell GPU, and 128 GB of RAM.

2.2.1. Patient cases

The prostate VMAT and brain PBS cases were selected as representative clinical scenarios with a high degree of conflict between robust target coverage and sparing of adjacent organs at risk (OARs). Such trade-offs are sensitive to geometric errors and therefore well suited for evaluating probabilistic planning methods. In addition, the inclusion of both photon VMAT and proton PBS cases demonstrates the versatility of the proposed framework across treatment modalities. Specifics for the cases were as follows:

Prostate case For the prostate case treated with VMAT, $\Sigma = 0.2$ cm and $\sigma = 0.3$ cm was used, chosen to be within the range of reported setup uncertainties (Cubillos Mesías *et al* 2016, Rudat *et al* 2016). The conventional plan used a margin of 0.7 cm according to the Van Herk rule $(2.5\Sigma + 0.7\sigma)$ (Van Herk *et al* 2000). The prescription was 77 Gy to the clinical target volume (CTV) delivered in 35 fractions.

The probabilistic optimization used a probabilistic constraint for the CTV prescription, and probabilistic objectives for the OAR clinical goals, as well as nominal objectives promoting a sharp dose falloff (see table S1 in the supplementary material for details). The conventional planning used the same optimization formulation, but with nominal functions instead of probabilistic ones, and with the margin-expanded target substituted for the CTV.

Brain case For the brain case treated with PBS, $\Sigma = 0.09$ cm and $\sigma = 0.12$ cm were used, chosen to be within the range of reported setup uncertainties (Cubillos Mesías *et al* 2016), as well as $\Sigma_r = 1.53\%$. The conventional plan considered systematic setup uncertainty of 0.34 cm, based on the margin recipe for 95% coverage probability $(2.79\Sigma + 0.7\sigma)$ (Van Herk *et al* 2000), as well as systematic range uncertainty of $1.96\Sigma_r = 3\%$ to also cover 95% of the range uncertainty and hence a total of $95\% \times 95\% \approx 90\%$ of the uncertainty space. The prescription was 60 Gy (RBE) delivered to the CTV in 30 fractions. However, as the CTV was directly adjacent to the brainstem, the main target clinical goal was relaxed to $D_{95} \geqslant 57$ Gy (RBE).

In the optimization, the probabilistic method used probabilistic constraints for the 'at most' clinical goals for the OARs and CTV, and probabilistic objectives for the CTV minimum dose clinical goals, as well as nominal objectives promoting a sharp dose falloff and low dose to the brain (see table S2 in the supplementary material for details). The conventional method used the same optimization formulation, but with worst-case robust functions instead of probabilistic ones.

3. Results

3.1. Prostate case (VMAT)

Probabilistic optimization consistently improved OAR sparing while maintaining target coverage (table 1). Over the ten OAR goals, the probabilistic plan increased the average fulfillment probability by 13.3 percentage points and reduced the P90 (90th percentile) OAR dose metrics by an average of 3.5 Gy compared to the conventional plan. The P90 of the mean external dose was reduced by 0.2 Gy. Nominal and average statistics are presented in table S3 in the supplementary material.

The dose–volume histograms (DVHs) in figure 2 illustrate these differences, showing lower OAR doses and maintained target coverage in the probabilistic plan and the dose distributions in figure 3

Table 1. Probabilistic evaluation statistics for the prostate case treated with VMAT: the percentage of passed clinical goals and percentile dose statistic values over the 1000 evaluation scenarios for the probabilistic and conventional margin-based optimization. For 'at least' goals, the 10th percentile (P10) values are shown, and for 'at most' goals, the 90th percentile (P90) values.

		Probabilistic			Conventional		
ROI	Clinical goal	Passed (%)	P10 (Gy)	P90 (Gy)	Passed (%)	P10 (Gy)	P90 (Gy)
CTV	$D_{100} \geqslant 77.0$ Gy	72	76.1	—	71	76.4	—
	$D_{99.8} \geqslant 76.9$ Gy	97	77.2	—	99	77.4	—
	$D_{99} \geqslant 76.2$ Gy	100	77.4	—	100	77.6	—
	$D_2 \leqslant 82.4$ Gy	100	—	81.4	100	—	81.6
Rectum	$D_{75} \leqslant 28.6$ Gy	23	—	31.3	1	—	33.6
	$D_{50} \leqslant 42.9$ Gy	98	—	41.0	72	—	45.0
	$D_{20} \leqslant 66.7$ Gy	80	—	69.0	50	—	73.2
	$\bar{D}$	—	—	46.8	—	—	46.8
Bladder	$D_{50} \leqslant 61.9$ Gy	93	—	60.3	78	—	64.7
	$D_{25} \leqslant 71.4$ Gy	32	—	76.8	11	—	77.2
	$D_{10} \leqslant 72.0$ Gy	0	—	79.2	0	—	78.7
	$\bar{D}$	—	—	56.5	—	—	56.5
Anal canal	$D_{60} \leqslant 33.3$ Gy	100	—	18.3	100	—	23.5
	$D_{40} \leqslant 38.1$ Gy	99	—	34.2	84	—	40.7
	$\bar{D} \leqslant 45.0$ Gy	100	—	28.9	100	—	34.4
Bulbus	$\bar{D} \leqslant 30.0$ Gy	93	—	28.3	89	—	31.3
External	$\bar{D}$	—	—	5.5	—	—	5.7
External−CTV	$\bar{D}$	—	—	5.3	—	—	5.5

show that it has slightly retracted all isodose lines between 50% and 100% from the rectum and towards the target.

The probabilistic optimization took 15 min for 130 iterations, and the probabilistic evaluation took 5 min. The conventional margin-based optimization took 1 min. The number of monitor units per fraction was similar between the plans: 439 for the probabilistic plan and 429 for the margin-based.

3.2. Brain case (PBS)

The probabilistic plan improved target coverage while maintaining or improving OAR sparing (table 2). For the CTV, the fulfillment probabilities increased substantially, from 19% to 84% for D_{95} and from 4% to 32% for D_{98}. Nominal and average statistics are presented in table S4 in the supplementary material.

For OARs, the most pronounced improvement was observed for the brainstem core maximum dose, where the passing probability increased from 2% to 56%. The probabilistic method also reduced the P90 levels for near-maximum doses for several structures, including reductions of 4 Gy (RBE) for the chiasm and 1.7 Gy (RBE) for the optic nerve. The P90 levels for mean doses were consistently reduced, with decreases of 2.2 Gy (RBE) for the brainstem surface, 3.9 Gy (RBE) for the brainstem core, 4.0 Gy (RBE) for the chiasm, and 0.6 Gy (RBE) for the brain. However, the P90 levels of the brainstem surface maximum and near-maximum dose goals increased by 0.4 Gy (RBE).

Figure 4 shows that the DVHs of the OARs of the probabilistic plan were generally lower than those of the conventional plan, and that the CTV coverage was higher. The dose distributions in figure 5 show that the probabilistic plan retracted the 90% isodose line away from the inside of the brainstem but had the 95% isodose line closer to it than the conventional plan. It also used the anterior beam more than the conventional plan, and had more heterogeneous nominal target dose.

The probabilistic optimization took 27 min for 200 iterations, and the probabilistic evaluation took 10 min. The conventional robust optimization took 1 min.

4. Discussion

For the two cases studied, the percentile-based probabilistic optimization produced plans with higher probabilities of clinical goal fulfillment and improved trade-offs between target coverage and OAR sparing compared to conventional planning. These improvements arise from the method's explicit control of the distribution of scenario doses over full treatment courses.

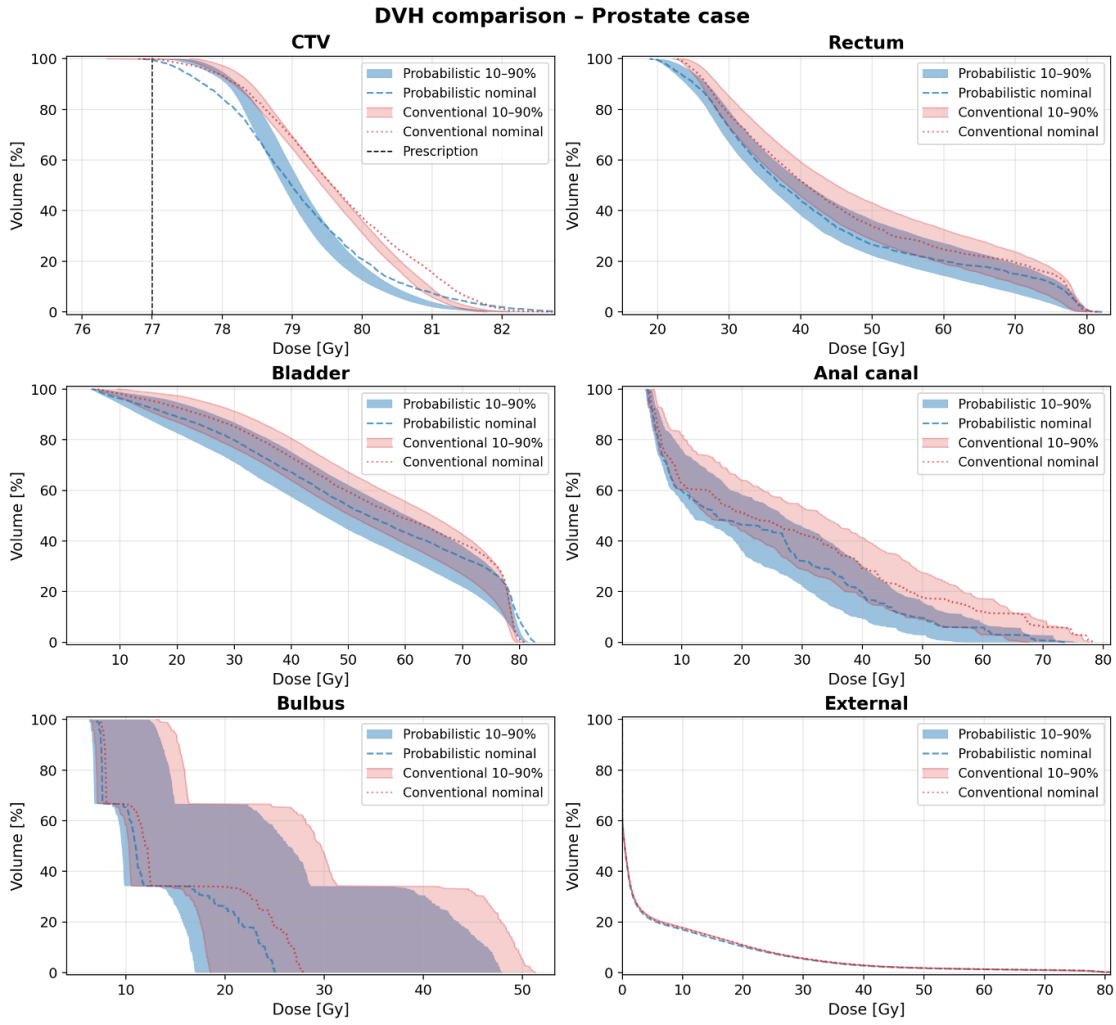


Figure 2. Nominal DVHs and 10–90th percentile DVH bands over the 1000 evaluation scenarios for the prostate case treated with VMAT, comparing the probabilistic method (blue) with conventional robust optimization (red).

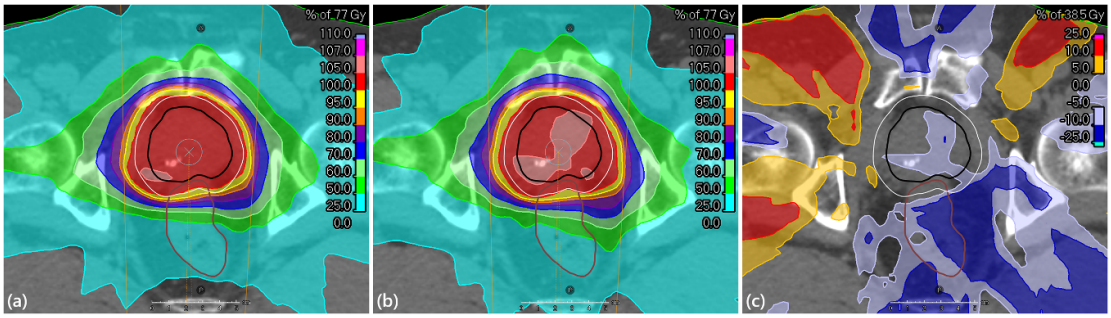


Figure 3. Transversal slices and dose distributions of the prostate case treated with VMAT for (a) the probabilistic method, (b) the conventional margin-based method, and (c) the dose difference (probabilistic−conventional). ROIs displayed are CTV (black), CTV + 0.7 cm margin (white), rectum (brown).

While this work focuses on probabilistic optimization, a natural first step toward probabilistic planning is the adoption of probabilistic evaluation. Applying probabilistic evaluation to conventionally optimized plans already enables a more informative assessment of clinical goal fulfillment probabilities than nominal or worst-case analysis, and can be introduced independently of changes to the optimization (Sterpin *et al* 2021, de Jong *et al* 2025).

Even when a probabilistic constraint is used during optimization, the achieved passing probabilities in the resulting plan may differ from the target values. In the prostate case, for example, the optimizer enforced a constraint aiming for a 90% probability of meeting the CTV minimum dose goal, which was

Table 2. Probabilistic evaluation statistics for the brain case treated with PBS: the percentage of passed clinical goals and percentile dose statistic values over the 1000 evaluation scenarios for the probabilistic and conventional robust optimization. All dose statistics are presented in RBE-weighted dose. For 'at least' goals, the 10th percentile (P10) values are shown, and for 'at most' goals, the 90th percentile (P90) values.

		Probabilistic			Conventional		
ROI	Clinical goal	Passed (%)	P10 (Gy)	P90 (Gy)	Passed (%)	P10 (Gy)	P90 (Gy)
CTV	$D_{98} \geqslant 55.8$ Gy	32	54.5	—	4	54.4	—
	$D_{95} \geqslant 57.0$ Gy	84	56.8	—	19	55.8	—
	$D_2 \leqslant 64.2$ Gy	95	—	64.1	93	—	64.1
Brainstem surf.	$D_{0.03\,\mathrm{cm}^3} \leqslant 56.0$ Gy	98	—	55.5	100	—	55.1
	$D_0 \leqslant 56.0$ Gy	62	—	56.9	56	—	56.5
	$\bar{D}$	—	—	16.7	—	—	18.9
Brainstem core	$D_{0.03\,\mathrm{cm}^3} \leqslant 54.0$ Gy	99	—	53.4	97	—	53.7
	$D_0 \leqslant 54.0$ Gy	56	—	54.6	2	—	54.6
	$\bar{D}$	—	—	16.2	—	—	20.1
Chiasm	$D_{0.03\,\mathrm{cm}^3} \leqslant 54.0$ Gy	100	—	41.6	100	—	45.6
	$\bar{D}$	—	—	17.9	—	—	21.8
Optic nerve (L)	$D_{0.03\,\mathrm{cm}^3} \leqslant 54.0$ Gy	100	—	47.8	100	—	49.5
	$\bar{D}$	—	—	21.5	—	—	22.8
Brain	$\bar{D}$	—	—	6.0	—	—	6.6
External	$\bar{D}$	—	—	1.8	—	—	1.9
External−CTV	$\bar{D}$	—	—	1.5	—	—	1.7

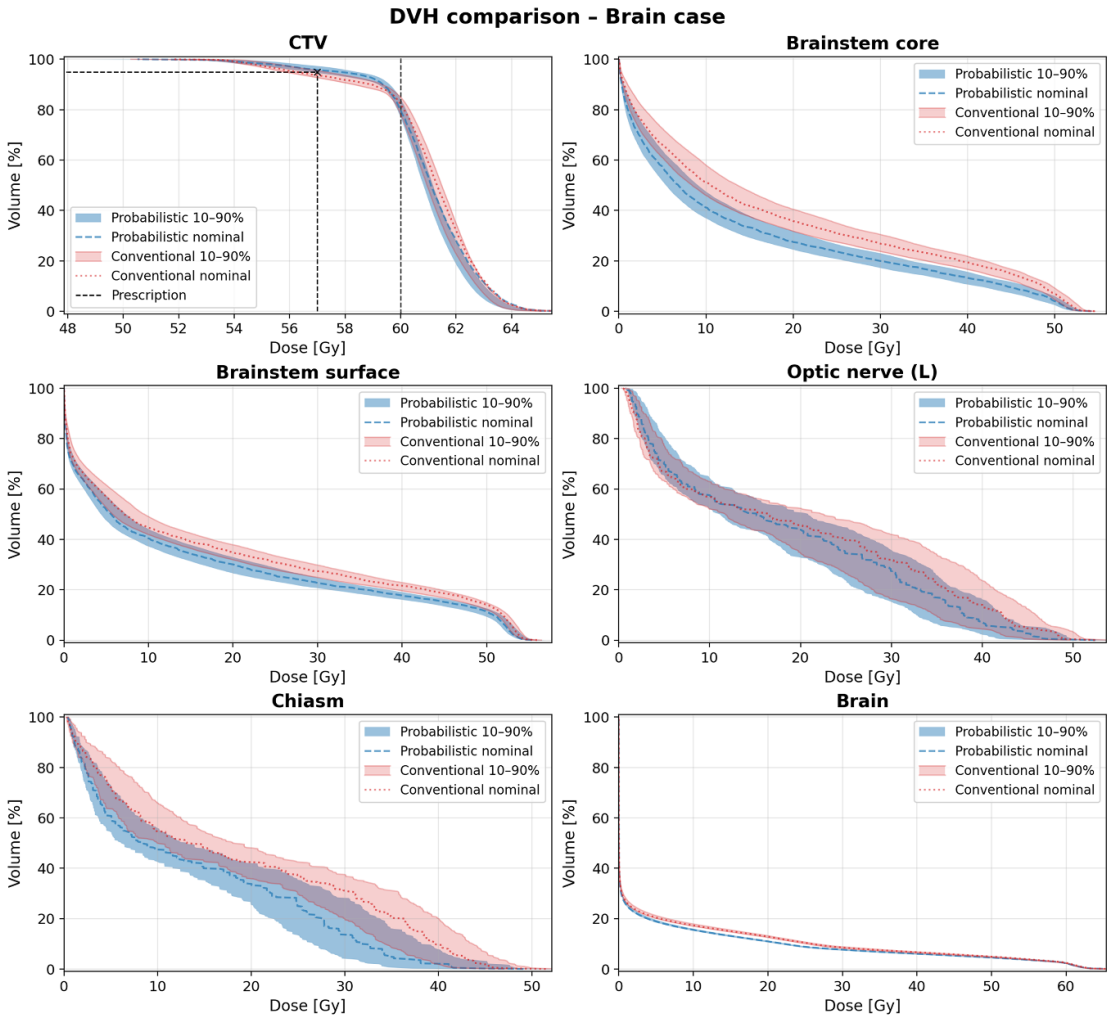


Figure 4. Nominal DVHs and 10–90th percentile DVH bands over the 1000 evaluation scenarios for the brain case treated with PBS, comparing the probabilistic method (blue) with conventional robust optimization (red).

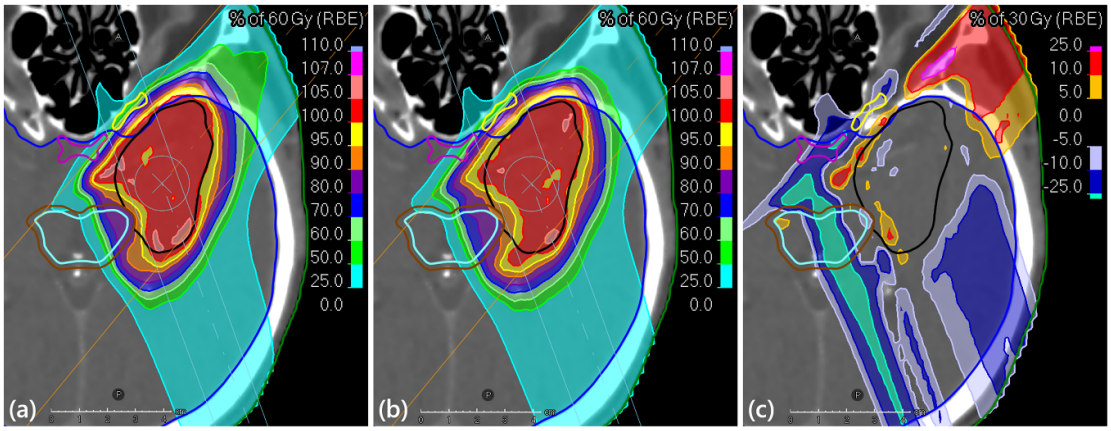


Figure 5. Transversal slices and dose washes of the brain case treated with PBS for (a) the probabilistic method, (b) the conventional robust optimization, and (c) the dose difference (probabilistic−conventional). ROIs displayed are CTV (black), brainstem core (cyan), brainstem surface (brown), optic nerve left (yellow), chiasm (magenta), brain (blue), external (green).

numerically satisfied to within 10^{-8}. Despite this, the resulting plan achieved only a 77% passing probability in the optimization scenarios and 72% in the independent evaluation. In contrast, the $D_{99.8}$ had passing rates of 97% in both optimization and evaluation. Such deviations can result from limited numbers of simulated treatment courses, constraint tolerances, differences between optimization and evaluation dose calculation algorithms (including the different resolutions of the setup shift grids), and the sensitivity of binary evaluation criteria to small dose variations.

Compared to previous probabilistic planning methods, the present framework introduces two main methodological extensions. First, both systematic and random errors are explicitly modeled without relying on dose blurring, enabled by a fast scenario dose approximation method. This extends applicability to photon treatments with pronounced density heterogeneities and to ion therapy. Second, the proposed percentile-trimmed power mean formulation allows smooth scaling between conditional expected value and percentile-based optimization while handling correlation between probabilistic goals.

Explicit modeling of random errors over full treatment courses enables future investigations of fractionation-dependent effects. For example, applications in hypofractionated or adaptive treatments and incorporation of radiobiological models, such as the linear-quadratic model, into the dose accumulation could be explored. For the approximate treatment course scenario dose calculation, alternative grids and interpolation schemes could be studied. In photon therapy for cases with limited heterogeneity in patient density, substantial speed ups could be achieved using the static dose cloud approximation (Witte *et al* 2007, Bohoslavsky *et al* 2013).

Percentile-based optimization is inherently non-convex and therefore does not guarantee convergence to a global optimum. Nonetheless, the percentile-trimmed power mean is closely related to DVH-based optimization functions, which have a long history of successful clinical use despite similar non-convexity. Convex surrogates such as CVaR generally do not yield optimal solutions to the percentile criterion, but can serve as initializations that are subsequently refined using the non-convex percentile formulation.

This study does not consider anatomical changes or distributional robustness. Simplified anatomical variations could be incorporated using interpolation-based approaches (Fredriksson *et al* 2021). However, combining anatomical changes with setup errors substantially increases the computational complexity. Robustness to uncertainty misspecification could be improved by the inclusion of random errors with different distributional assumptions in both the optimization and evaluation.

5. Conclusion

We have presented a probabilistic optimization framework for robust radiation therapy treatment planning that explicitly controls the probability of goal fulfillment, and evaluated the framework using both VMAT and proton PBS planning. The percentile-trimmed power mean provides an interpretable risk measure spanning expected value conditioned on a percentile and direct percentile optimization. Combined with efficient approximate dose calculation, the method enables realistic simulation and optimization of fractionated treatment courses under uncertainty. The example cases show that the method can yield improved probabilistic trade-offs between target coverage and OAR sparing compared to conventional planning methods.

Data availability statement

The data cannot be made publicly available upon publication because they are owned by a third party and the terms of use prevent public distribution. The data that support the findings of this study are available upon reasonable request from the authors.

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